Omeed Djassemi

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EDUCATION

California Polytechnic State University- San Luis Obispo, B.S.
Major: *Biomedical Engineering*
Dean's List: Winter 2019, Spring 2019, Fall 2019, Winter 2020
University of California – San Diego, PhD (anticipated) *GPA: 3.9*
Major: *Bioengineering*

WORK & ENGINEERING EXPERIENCE

UCSD Department of Bioengineering
Graduate Student Researcher – Joseph Wang Laboratory for Nanobioelectronics **8/2022 – Present**

- Designed and manufactured neonatal Pacifier capable of performing metabolic and biophysical measurements.
- Developed integrated potentiostat hardware & algorithms for biosensors, optical blood pressure, HR, and SpO_2.
- Participated in and published human lactate microneedle sensor studies in an intensive care unit (ICU).
- Fabricated stretchable microneedles for Na^+, K^+, Lactate, Glucose monitoring in interstitial fluid.

Electrical and Computer Engineering Makerspace Lab Technician **9/2021- 6/2022**
Teaching Assistant – CHEM 6C, BENG 110, BENG 226, BENG 172 **3/2022 – 6/2023**

- Bioengineering Department TA of the year 2022-2023.

Dexcom – *The industry standard in continuous glucose monitoring* **6/2022-8/2022**
Project Management & Systems Engineering Intern

- Created algorithms for increasing the precision of estimated battery life of a manufacturing R&D instrument.

Cal Poly Department of Biomedical Engineering **3/2020-7/2021**
Researcher – Micro/Paper-Fluidics (subject to NDA)

- Lateral Flow assay design for stroke detection through COMSOL computational fluid dynamics.
- 3D Printing Specialist. Responsible for a lab of seven 3D Printers. Supported multiple research teams.

Cal Poly Department of Biomedical Engineering **1/2020-6/2021**
Biomechanics Instructional Student Assistant

- Instructed and prepared Biomechanics laboratories with ECG, Respiratory, and Anthropomorphic instruments.
- Assisted the Professor in grading and laboratory tasks in addition to a significant course load and work.

Cepheid - *The industry standard for innovation in automated molecular diagnostics* **6/2020-9/2020**
Process Engineering Intern – Sustainment Focus

- Assessed and troubleshooted cleanroom automation lines to increase medical device yield for the global pandemic.
- Designed and researched experiments to mitigate the risks associated with the reduction in inspection of a medical device.

Cal Poly College of Engineering: *Interdisciplinary Senior Project* **9/2019-6/2020**
Speech Pathology Robot

- Design challenge involves designing a closed-loop robotic system capable of providing visual and audio feedback.
- System uses voice recognition software and Bluetooth communication to operate autonomously.

OptiScan Biomedical Corporation- *Developer of novel CGM intended for Surgical ICU* **6/2019-9/2019**
R&D Engineering Intern

- Designed and programmed a cleanroom manufacturing fixture that measures air gaps of microfluidic component(s).
- Integrated stepper motor and laser displacement sensor in manufacturing fixture to automate and decrease Validation time.

SKILLS & QUALIFICATIONS

- Wearable Electrochemical Biosensor R&D
- Vital Sign instrumentation R&D
- Altium for bioinstrumentation design
- SU-8 Microneedle fabrication and design
- SEM, Thin film deposition, Photolithography
- SolidWorks
- MATLAB
- Arduino/STM/Python IDE
- Electrochemistry
- Fluid Transport Theory & Phenomena
- Altium (PCB Design)
- Advanced 3D Printing and Maintenance
- Bioengineering TA of the year 2022-2023
- Instrumentalist: Guitar, Bass, Drums, Harmonica

PUBLICATIONS (~550 CITATIONS; H-INDEX 10; *FIRST AUTHORED)

Google Scholar Profile

[1] Redox Cycling Amplified Electrochemical Lateral-Flow Immunoassay: Towards Decentralized Sensitive Insulin Detection, ACS Sensors, https://doi.org/10.1021/acssensors.3c01445

[2] Continuous Interstitial Fluid Ketone Monitoring via Wearable Microneedle Patch Platform, ACS Sensors, https://doi.org/10.1021/acssensors.3c02677

[3] Touch-Based Potentiometric Sensors for Simultaneous Detection of Urea and Ammonium from Fingertip Sweat, Sensors & Actuators B: Chemical, http://dx.doi.org/10.2139/ssrn.4748241

[4] Integrated fingertip wearable microgrid system for energy management and sensing, Nature Electronics, https://doi.org/10.1038/s41928-024-01236-7

[5] A Passive Perspiration Based Wearable Platform for Continuous Sweat Glucose Monitoring, Advanced Science, https://doi.org/10.1002/advs.202405518

*[6] Simultaneous and Rapid Detection of Glucose and Insulin: Coupling Enzymatic and Aptamer-Based Assays, Analytical Chemistry, https://doi.org/10.1021/acs.analchem.4c04289

[7] Earable Multimodal Sensing and Stimulation: A Prospective Towards Unobtrusive Closed-Loop Biofeedback, 10.1109/RBME.2024.3508713

*[8] A Touch Enabled Hemodynamic and Metabolic Monitor, Advanced Science, https://doi.org/10.1002/advs.202502138

[9] Smart closed-loop drug delivery systems, Nature Reviews Bioengineering, https://doi.org/10.1038/s44222-025-00328-z

*[10] Monitoring blood lactate dynamics through sweat and interstitial fluid biofluids, Talanta, https://doi.org/10.1016/j.talanta.2025.128985

*[11] Clinical Evaluation of Microneedle Biosensors for Continuous Lactate Monitoring in Critically Ill Patients, ACS Sensors, https://doi/10.1021/acssensors.5c03699

*[12] Continuous Lactate Monitoring for Real-Time Fatigue Assessment in Baseball Pitchers, Biosensors & Bioelectronics, https://doi.org/10.1016/j.bios.2026.118611

[13] From material innovation to clinical translations: Interface stability and AI-enabled wearable health tracking, Next Bioengineering, https://doi.org/10.1016/j.nxbio.2026.100038

[14] A fully integrated smart ring for daily biochemical monitoring, Nature Communications, https://doi.org/10.1038/s41467-026-75980-z